Ep. 368: Richard Clarida on the Warsh Fed, the AI Capex Boom, and Global Economic Fragmentation

Podcast: Macro hive

AI Executive Summary

Core Thesis The episode explores the resilience of the global economy during a period of significant structural transition (circa 2025â€“2026), characterized by the rise of AI-driven capital expenditure and increased geopolitical fragmentation. Richard Claridge argues that while fears of a massive recession or 1970s-style wage-push inflation have not materialized, the economy is undergoing a profound shift where income is moving from labor to capital, and monetary policy must adapt to new inflationary pressures stemming from supply-side shocks (such as tariffs and AI-driven demand for energy and chips) rather than traditional labor market overheating.

Key Arguments & Data * Economic Resilience vs. Recession Fears: Despite widespread predictions of a global recession in 2025, US growth has remained resilient at approximately 1.5% on average. However, the "AI boom" is not a pure GDP driver because much of the CapEx "leaks" out of the US via imports for data centers and may "crowd out" spending in non-hyperscaler sectors. * **The Shift from Labor to Capital:** A secular trend is evident where the labor share of national income is at its lowest level in 80 years. Unlike the 1970s, current inflation is not driven by "cost-push" wage inflation; instead, high profit margins and corporate growth are absorbing much of the economic gains. * **Inflationary Pressures in the AI Era:** While recent CPI/PPI prints have been soft, new upward pressures are emerging from the AI boomâ€”specifically higher electricity prices and increased costs for memory chips (estimated by PIMCO to add roughly 0.5% to core PCE). Additionally, tariffs remain a significant risk to reaching the Fed's 2% target. * **The "Warsh Fed" and Monetary Policy:** The Federal Reserve committee under Chair Warsh has shifted toward a more hawkish stance compared to previous projections. Claridge suggests that the Fed may increasingly use its balance sheet (size and maturity composition) as a tightening tool, rather than relying solely on adjusting the federal funds rate. * **Critique of Forward Guidance:** Claridge argues that "overly prescriptive" forward guidance can be counterproductive by creating difficult "off-ramps" for central banks. He suggests that when interest rates are well above the zero bound, the necessity and utility of explicit forward

guidance diminish. * **Geopolitical Fragmentation:** The era of hyper-globalization has ended, replaced by a "rupture" in global security, trade, and financial alliances. However, Clarargidge maintains that capitalist economies possess a high degree of adaptability to these new fragmented realities.

Actionable Takeaways * For Investors (Sector Rotation): Look beyond the primary AI software/chip companies (the "hyperscalers") to the "old economy" beneficiaries of the AI build-out, such as construction and heavy machinery (e.g., Caterpillar) driven by data center infrastructure demand. * **For Macro Strategists:** Monitor the Fed's balance sheet maneuvers and the cost of inputs (energy, memory chips, and tariff-impacted goods) rather than focusing exclusively on labor market tightness or interest rate hikes, as the latter is no longer the primary inflation driver. * **For Career Development (Young Professionals):** Rather than fearing AI substitution, focus on skills that *complement* AI technology. Additionally, consider vocational/skilled trade paths (e.g., electricians, plumbers), which are currently experiencing a structural shortage and benefit from new corporate-backed apprenticeship programs like the "Meta Job Academy." * **For Global Business Planning:** Incorporate higher levels of "ex-ante" geopolitical risk into supply chain and alliance strategies, acknowledging that the world is moving toward fragmentation rather than integration.

Full Transcript

Welcome to the MacroHive Conversations podcast. I'm Bilal Hafiz, head of strategy at MacroHive. Before we dive in, a quick update. Following MacroHive's acquisition by the BGC Group, BGC has launched a new regulated business called MacroHive Markets. For the first time, onboarded clients can also access execution services via MacroHive Markets. This new workflow will allow macro institutional investors to benefit from strategy and quant insights, alongside access to global liquidity. Please contact Andrew Simon or Gail Mojibriyan

Bloomberg or click the link in the description below for more details on McRide markets and its service offering. Now onto this episode's guest, Richard Clariger. Richard is a managing director and global economic advisor at PIMCO. Prior to rejoining PIMCO in 2022, he was the firm's global strategic advisor from 2006 to 2018. He served as vice chairman of the board of governors of the U.S. Federal Reserve System from September 2018 to January 2022. Richard is also professor of economics and international affairs at Columbia University. Prior to joining PIMCO in 2006, he was assistant secretary of the Treasury for Economic Policy, in which he served as chief economic advisor to two U.S. Treasury secretaries. Now, on to our conversation. Greetings and welcome, Richard. It's fantastic to have you on the podcast. Looking forward to it. Great. Now, obviously you're kind of well-known amongst our audience, but I did want to get our audience to know something about your origin story. So, you know, what did you study at university? Was it inevitable you'd end up in finance, econ, policy? Okay. Well, quick origin story. I grew up in downstate Illinois in a little coal mining town with about 8,000 people.

was lucky to have very educated parents, and so valued that, went to the University of Illinois, the flagship state university in the 1970s, studied economics, fell in love with it. Actually, when I took freshman economics with 800 folks in a big lecture hall, and so I was an econ major, I was very fortunate to have a mentor, a professor who sort of put me on a path of what I would need to do to apply to to graduate school and did that. And in those days, unlike today, you could go right from college to graduate school. So

at the age of 21, and I hopped in my beat-up car and drove from Champaign, Illinois to Cambridge, Massachusetts, went to Harvard, studied macro, and so that's the origin. Yeah.

That's fantastic. And of course, you've had a quite illustrious career since then. Now, obviously, the big topic at the moment is the Fed, inflation, all these kind of big macro topics. But I guess perhaps we can start with, have you been surprised about the resilience of the US economy this year? So some people started there saying growth is too narrow, just AI. And then we had the Iran war. So people thought that will derail it. But it's been much more resilient. I would actually go a little bit further. I think the resilience story was an important element of 2025 that got largely ignored by the financial press. To be blunt, much of the financial press on Liberation Day declared at the end of capitalism, global recession inevitable, super high inflation in the US. If you actually look at the outcome in 2025, you'll see the global growth came in exactly as in 24. You'll see that US inflation did not go up. It didn't go down. It stayed around 2.8 or 2.9%. And so I think the resilient story was a story not just for the US, but the global economy in 2025.

2026, obviously we're only about six months into the year. Clearly the US is benefiting substantially from an unprecedented AI CapEx boom. However, arithmetically it's not the same as translating into GDP for a couple of reasons. One, a lot of the inputs for the AI CapEx boom are imported. So there's leakage through imports, good for the rest of the world, especially if you sell stuff that you need for data centers. And secondly, there is some crowding out in the data. If you look at non-hyperscalar capex spending, not only is it soft, it's actually by some measures been contracting. Also remember, we did have a fiscal boost this year in the US because the one big beautiful bill kicked in. And so at least through, if you look at the last quarter of 25 and the first quarter of this year, growth I think was probably about one and a half percent on average. We'll get the Q2 numbers later on this month and it'll be interesting to see where those land.

No, that all makes sense. And so I mean, before we get on to inflation, I did want to ask what your thoughts on the labor market, you know, because that's one of the things that has thrown a lot of economists off because we

have not only some demand factors, supply factors as well. And so there's this, you know, sense of like, what is the equilibrium payrolls that we need to look at? Really an excellent question. And I think an important element of why the power fed cut rates last year with inflation above target, they were concerned about the labor market. Jay Powell, several times made reference to what he called, quote, a curious balance in the labor market. But the basic idea that you highlighted that employment growth was incredibly sluggish. Indeed, if you just plot, say a six or 12 month average of employment growth, it basically collapsed from 150 to 200 K a month down to basically zero last year. But of course labor supply growth also slowed because of a change in the enforcement of immigration laws. And so I think the Fed's concern last year was to interpret soft labor employment growth as an indication of weak labor demand with the benefit of hindsight. A couple of things have happened. One is we've seen some pickup in employment growth this year, which is probably welcome. But importantly, we've actually continued to see that wage inflation adjusted for productivity.

has continued to come down. So I think the important point is from the point of view of both the outgoing pal and the incoming wash bed, unlike in past cycles, the labor market is really not. a driver of inflation right now. In fact, let me get on my soapbox a little bit here if I can. I get tired of all the analogies to the 1970s in the US. In particular, I was a college student in the 70s and so sort of was navigating economics at that time. And of course, you did have oil shocks and you did have high inflation, although it was in the double digits. But a very important difference between this cycle and the 70s is that in the 70s, you had not only high price inflation, you had very high wage inflation. You had what economists call a cost push cycle in the labor market, which was driving up price inflation. We don't see that at all now. In fact, if you look at wage inflation adjusted for productivity, it's running right where it was back in 2018, 2019. So I think.

you know, the Warsh Fed is focusing on getting down to 2% inflation. But I also hope that they diagnose that the problem right now is, you know, that there aren't too many people working. I think there are other sources of inflation pressure, but it's not the labor market. And just on the labor market, you know, why aren't wages growing more strongly? The unemployment rate is incredibly low level. And, you know, there's, you

know, labor supply has gone down. So presumably that should up wages growth seems strong. So why aren't wages stronger? It's a great point because if you rewind the proverbial tape back to 2024, there was a lot of hand wringing about if there is a crackdown and immigration enforcement is elevated, then it will create a problem for the Fed because it'll push up wage inflation because you won't have enough workers. We didn't get that. And so even though ex ante that could have been the outcome, we've basically seen a downshift in labor demand commensurate with labor supply. You know, there's also a school of thought. I haven't necessarily signed up to it, but I'm aware of it, which makes the case that perhaps one of the spillover implications of AI or concerns about AI in the service sector, why call our service sector, is the fear of AI and uncertainty about it.

may increase the relative bargaining power of employers over employees. And so I think right now that's anecdotal, but certainly even though we have a historically low unemployment rate, we just do not have wage inflation as a driver of price inflation right now. And you know, another school of thought. around this wage issue is that, at the same time as low wage growth, you're seeing incredibly high profit margins and profit growth. So is there kind of an issue around monopolistic behaviors in the US in some ways that you're seeing growth just occurring to corporates rather than to workers? Well, the short answer is yes. If you look at calculations of national income and you divide it into capital income and labor, Income by by many of those calculations labor share of national income is now the lowest it's ever been in the data going back eighty years not always some measurement issues but the trend line is pretty clear and certainly those estimates are consistent with other estimates that we see for example.

One of the stories in the equity markets have been not just excitement about the potential of AI, but profit growth has been very strong. Indeed, I think I'm not really an equity guy, but I talked to them. One of the stories about the very, very well-supported US equity market is not just a hyperscaler story, but profits broadly in the S&P, surprised on the upside. And this is obviously, very important frankly it's something that the economics profession has not done a very good job of trying to understand or engineer so there's been some work on for example the china shock that economists

looked at in terms of manufacturing employment and profitability but the reality of the data i think it is unmistakable.

There has been a super secular trend away from labor towards capital and that has continued and it's certainly a relevant part of the outlook. And let's move on to inflation. I mean, this will come out, you know, about a week after the CPI sort of print, but we had an incredibly weak, low CPI print. I mean, first of all, any comments about the very recent print and then more generally about the overall trend in inflation? I think. you know, from the point of view of the Fed and the economy and markets, it was welcome. I don't think it was totally surprising because we knew that energy prices in June were lower than May. So month over month, we would get that headline decline. I think what was welcome is that the core measure also came in softer than expected. And importantly, there were visions to previous months in the downward direction. We also got a very, very pleasant surprise in terms of the PPI.

producer price index numbers that came out a day later. And those also came in on the soft side. And so, as Chairman Warsh indicated, one good data print does not represent mission accomplished for the Fed, but certainly welcome. And then, of course, we'll get the PCE price index data at the end of the month. But people I respect here at PIMCA who do that calculation based upon the numbers we have, think that that number will come in somewhere below 0.2 on a monthly basis, which again would be welcome. And in terms of the overall level of inflation of the last couple of years, obviously, you know, post COVID post the Fed hikes, we saw inflation fall. But, you know, it hasn't gone back to two. It's stubbornly high core headline, whatever, around the three-ish mark. I mean, why do you think it's so hard to get back down to two? Well, let's start with an important observation, which is that other central banks have been able to do that. So before the hostilities in the Middle East, inflation in the Eurozone was running at two, inflation in Canada was running at two. So even though there was a common shock and everybody had inflation that was way too high in 2021 and 2022, And even though globally central banks were able to navigate back towards their target, you had central banks that actually got there. And so in some ways, why was the Fed, even before the tariffs and the oil shock, why was the Fed unable to get inflation all the way back to

two? Now, it got pretty close. I think on a year over year basis, a core got down to two four in September of 2024. And in particular, housing service inflation has continued to drift down and I think the Fed can make a plausible case which they've made that but for the effect of tariffs in 2025 inflation at the end of 2025 would have been pretty close to two. So I think the Fed the Powell Fed's thinking is they never declared mission accomplished but they were on a path to that but for the tariffs, you know, we will never know, but that's not an implausible argument. However, I think the challenge right now in 2026 is the upper pressure on inflation, while it's not coming from the labor market, is at least for some time has been a little becoming from the AI, CapEx, boom, and in particular, the headline basis pushing up electricity prices on a core basis.

pushing up the price of memory chips. So, you know, memories, you know, microwaves thought it was like computer memory is sort of being a commodity. And which if you sort of look at the share price of memory companies is what we saw. But whenever you've got a rapid rise in demand and a relatively inelastic supply curve, you have higher prices for memory, which we do. And you have, and then that gets passed through into iPhone prices and software prices have also gone up and so we've done some calculations at PIMCA that are consistent with what I've seen elsewhere that this year alone the direct effect on the PCE price index of memory, higher memory and software prices will probably add about half a percentage point to core inflation and so I think the challenge for the Fed and I think we've picked up on this in some commentary from my good friend Governor Chris Waller the other day that you know, after five going on six years of missing the inflation target, even if you can convincingly argue a non-monetary reason why inflation has been above target, you know, at some point, the fat has to take ownership for that miss. And of course, that's a potential that creates a challenge if it's a supply-driven shock. So I think three things. As I said, I think by 2024, an underlying inflation was certainly below two and a half.

And if you make a calculation for the direct effect of tariffs and memory chips would probably be around two right now, but that's not the world that we're in. That's another world that we weren't in. And so I think the Warsh Fed will have to navigate this world as it takes it. I mean, we've mentioned

Warsh a few times now. you know, obviously there's so much conversation around the Warsh Fed, you know, one in terms of his relationship with the president, another in relation to historic comments that sort of implied he's hawkish, but then, you know, the president sort of picked him. So maybe he's dovish. And then also what we've seen more recently in terms of the way he's managing the Fed, you know, family fights, you know, less, you know, just it's a different dynamic dropping forward guidance. You know, what's your sense of the Warsh Fed?

Well, obviously, an important topic, and we can't cover it fully now. I'd make a couple of observations. First, let's look at where the FOMC that Washington Harris, the committee that Washington Harris, let's look at where that committee was in March. That was four months ago. In March of this year, after hostilities had commenced in the Middle East and oil prices had gone to 100 a barrel, the Fed officials submitted their projections and their dots. And what they showed is that No one wrote down a rate hike for 2026 and 12 people wrote down a rate cut. And the Fed was projecting core inflation of around 27. You fast forward to Warsaw's first meeting, which was in June, only one person wrote down a rate cut and nine people wrote down a rate hike. So that's the same committee. And so the way I like to think about it is because monetary policy is made by a committee, the chair only has one vote, You know, May and others call for the Warsh Fed in March was really a call on the committee he was inheriting, which in March wanted to cut rates. Now, the committee he inherits because of the inflation forecast being revised up does not want to cut and many want to hike. And so I think that's what you have to understand. He's inheriting a committee that does not want to cut. He's inheriting committee that wants to hike. And he has very, very consistently said he doesn't want to tip his hand.

other than to emphasize his goal and the committee's goal of getting inflation back down to target. And of course, he's also announced not one, not two, but five distinct task forces. And he's also said that they will be providing their recommendations at the end of this year. And so I think, especially with these benign inflation readings that we got recently, the best bet is that going into the July meeting in a couple of weeks, and there will not be projections released at that meeting, that there will be a discussion, but we would not expect that the committee to pull the trigger on a rate

hike. And whether or not we get a rate hike in September, November, or December is to some extent going to depend on the flow of the data. Maybe a final word, a challenge, PIMCO's baseline view is that we will not get a hike out of the Warsh Fed this year. We don't think we're going to get a cut either.

And we think that if the inflation data begins to return back towards 2% next year that that will give the committee room to cut rates perhaps once or twice. But very much a data dependent forecast. I think that an important issue is that Chairman Warsh has indicated that he's not a big fan of forward guidance. He thinks the Fed talks too much. There's not really much he can do about that because individual Reserve Bank presidents and members of the Federal Reserve Board, you know, have a First Amendment right to go on CNBC or Bloomberg television or give speeches. And so we're not going to be entering a world of no or even less fed communication. The real question is how will the chairman communicate in that world because other folks are going to be talking. And right now he's really emphasized goals and he's emphasized the toolkit. And One of the things I learned from his testimony on Capitol Hill this week is that he very much is sticking with a view that he articulated before he became chairman, which is that he thinks the Fed should view the balance sheet as well as rates.

as part of the toolkit. And he's broadly hinted that the balance sheet could be used to tighten financial conditions on its own without necessarily adjusting rates. And so down the road, I think Fed watchers are going to have to at least be alert to the possibility that if the Fed needs to tighten, it might not necessarily be raising the federal funds rate. It might be reducing the size of the balance sheet or the maturity composition of the balance sheet. and unfold guidance. I mean, what's your view, you know, on forward guidance or not? Because for so many years now, we were told by central bankers, not just in the US, but many of the central banks are in the world. Yeah, I've had Mervyn King on the podcast before, for example, you know, back in the day. Lord Mervyn King. Lord Mervyn King. Yep, correct. You know, he, you know, to West fan charts and providing a lot of guidance. So that kind of became the norm. And so dropping it, you know, seems quite a big deal. I mean, what's your take on this? All right. So I'll start with my conclusion and then give a little background. So I actually thought before I

joined the Fed and I gave some speeches that made this point that for guidance and quantitative easing, both will continue to be parts of the central bank toolkit. However, I don't think that these new tools of monetary policy are exempt from the laws of economics. I think returns can diminish. I think there may be benefits, but there may be costs.

In particular, in a world in which the central bank is not constrained at the zero bound, the necessity for forward guidance is less. Obviously, if you're in a situation like the global financial crisis or like the power fed was in in 2020, when your economy's in collapse and you've cut rates to zero, you don't have a lot of the traditional toolkit. You have no other traditional toolkit available. So you have to go to plan B. But away from the zero bound, I think in many circumstances, including currently, both John Williams and Chris Waller this week made this point that given the uncertainty in the environment, the value of forward guidance at this juncture is less. So I think I would not, if I were back at the Fed's sale, I'll never use forward guidance, but I think you want to think long and hard about how you do it. In particular, the details matter. And I've made the point since I left the Fed that in retrospect, the forward guidance that the Fed offered in September of 2020 that it would not raise rates until both inflation had returned to target and the labor market had returned to full employment did not really pass a cost-benefit test. And in particular, also, that statement did provide the Fed an off-ramp saying that the policy was in service to keeping inflation expectations well anchored and thus the Fed could deviate from that guidance if that was a concern.

And so I think overly prescriptive forward guidance can in practice prove more difficult than the textbook models that I and others have written down would would indicate it. So I guess bottom line is on forward guidance. There's a time and place where it's valuable. I would never say never. But certainly I would have a view that doing a pretty consistent and perhaps controversial. cost-benefit analysis up front would pay good dividends. Of course, you can make a case that both the Bernanke Fed in 2008 and 2009 and the Pal Fed in 2020 in the fog of economic collapse, financial dislocation in the case of the Pal Fed. It's important to remember that in March and April of 2020, we basically shut down the global economy. 22 million Americans lost their jobs in two months.

GDP collapsed at a 30% annual pace. And so certainly in retrospect, the Fed and other central banks were not finally calibrating down to the basis point or down to the billions of dollars balance sheet and interest rate decisions. But I think a fair criticism is once the economy began to recover, central banks stayed too long. with quantitative easing and forward guidance. And so I think there needs to be an exit strategy as well as a commitment to use the tools in a downturn. And on the potential Fed reforms with all these working groups, of course, when you were at the Fed, you know, there was also like a review. Yeah. What are the looking back and then at your experience of being part of that and then looking at the wash fed now? I mean, well, any comments or is this the right way of doing it or what the pros and cons of this approach. Sure. So maybe a little bit of institutional perspective. You know, many prominent central banks have periodic reviews of their policy framework. The Bank of Canada, for example, does a monetary policy review every five years. The Norges Bank in Norway, and I've advised that does a review either every year, every two years.

under both Andrew Bailey and Mark Carney, the Bank of England undertook reviews. And so it was, I think it was and is appropriate for the Fed to have a similar process. And one of the things the Pal Fed did in 2020 is said that the committee anticipates it'll make sense to do a review every five years. And so an important outcome of the Fed's 2025 review was to recognize the reality that the world in 2025 looked a lot different than the world in 2020. 2020, the Fed had not been trapped at the zero bound inflation expectations were not below 2%. And so in a world in which the zero bound is not a constraint and missing the inflation target on the downside is not a problem, the Fed could just return to traditional symmetric inflation targeting. And I think that makes sense. The Fed's review in 2025 also left open the door that if future circumstances were to be similar to what the Fed encountered between 2008 and 2020. Remember at some point, rates were negative in the Eurozone Japan. At some point I think there was close to 20 trillion euros of sovereign debt in Europe that traded at negative rates. Folks were talking about secular stagnation and interest rates being below growth rates. And so to some extent you want policy to be responsive to the world in which you're operating, but it's also important then to be responsible if the world in which you're operating

changes to adjust the framework. And so I think where the PALFED landed a year ago now makes sense. And I really would not expect right now the Warsh Fed to change that basic inflation targeting framework very much if at all. In fact, they've reaffirmed the 2% target. And so I think the way in which the Warsh Fed deploys the toolkit forward guidance and quantitative easing may well be somewhat different than we've seen before. I think the underlying framework in place is probably one that we'll endure. And one of the working groups is the impact of productivity or impact on AI productivity and so on. And so what's your take on that? Well, at PIMCO, one of my jobs is to organize an annual internal set of meetings, which we call our Secular Forum.

something we've been doing for more than 40 years. And not surprisingly this year, we had several sessions on AI in fact, including some sessions in advance we did by Zoom. I think we had five distinct sessions on AI. So I wouldn't pretend to be an expert, but I've certainly been reading and hearing from experts. And so I would say a couple of things. First is, I think there is no doubt here or among most people that AI will eventually increase the productivity of the economy in the simple sense that for any given labor force will produce more output. There is a wide range of estimates of how big that productivity boost will be. On the one hand, you have some hyperscaler firms. Anthropic comes to mind where they're predicting a doubling of productivity growth. So say 2% a year to 4% a year. At the other end of the scale, you have of Nobel laureate MIT economist Duran Asimoglu, who in a paper about a year or two ago projected that the boost from AI might be less than one percentage point in total over 10 years. I think it's fair to say there's a wide range of estimates, but there's somewhere between...

1% over 10 years and 20% over 10. But I think the direction is clear. This is a transformational general purpose technology, and it will make us and other countries more productive. The big question is to get from here to there, wherever there is, is going to require a massive amount of capital spending. And so although the ultimate increase in supply may develop and emerge, in order to get that higher supply, you need to increase aggregate demand today. And so there's an issue about the transition from here to there. And even now Chairman Warsh acknowledges that that is and will

continue for some time to put upward pressure on prices, which potentially could be a challenge for monetary policy. I think probably an even deeper potential challenge for the economy is what are the implications of this transition period?

to technological adoption of AI for the labor market, because you can get higher productivity with the same level of employment, but just more output per worker. Or you can also get productivity with higher output and fewer workers. Now, I think it's pretty clear, economic history is clear that in all past historical examples of the introduction of transformational general purpose technology, eventually there is no cruel trade-off. You get both more productivity and more more jobs. The US economy is incredibly more productive today than it was 30 or 40 years ago, but employment's also higher. And so I think most economists and I would have confidence that in some undefined long run, we'll be both more productive and have the same or higher levels of employment.

But certainly there are scenarios, perhaps not implausible scenarios, in which that transition is bumpy for particular sectors or occupations. I mean, right now, we have not seen it. And also remember that in order to build out the data centers that we discussed, You know it creates a lot of demand for old economy 20th century technology unique construction workers bricks and mortar unique caterpillar tractors to build data centers i am sharing anecdote with you and your listeners i was doing television a couple months ago. And the host on live television asked me about ask me a similar question i gave a similar answer about bricks and mortar and caterpillar tractors and on live television she said well rich did you know that caterpillar.

is a 900 stock. And I said, I had no idea. But if you look at a chart of the price of counterpillar stock, I think it's quadrupled in the last couple of years. It looks like a tech stock. One of the best ways exposed to have traded the AI boom was not NVIDIA, but was counterpillar share price. Absolutely. And if we kind of bring this back to monetary policy, then is it useful to think about this in terms of our star, whether we should push up our star or not? Or how should the central banker think about this in that context? Yeah. Well, as a matter of economic theory, and again, you never

want to end with economic theory, but it's useful at least to consider. So a matter of economic theory, the answer is yes. Economic theory says other things equal.

faster trend growth rate pushes up equilibrium interest rates. That's also what economic historical data says. And so I think the, put it this way, I don't know the effect of AI on interest rates, but if you tell me AI is going to boost productivity growth from 2% a year to 4% a year, real interest rates are going to be higher. So the uncertainty is not so much about the direction of travel. It's how big is going to be the ultimate productivity of boom. Now there's a related but a subtle point I'd like to also introduce the conversation, which is that the horizon also matters. I mean, no one would dispute that connecting the global economy via high speed internet and giving everybody on his desk or in his pocket a computer has boosted productivity.

But the effect on productivity growth was only about seven years. If you look at productivity growth in the US, it began to pick up in the mid 90s. It went from below two to above 3% by the mid 2000s. And then it went back below 1%. And so a subtle issue is the level effect of higher productivity birth the growth effect. And it's only the growth effect that should drive the real rate. So if you just tell me that in five years GDP per worker will be up by 7%, but beyond that growth returns to its prior pace, that might not have an effect on real interest rates. And so sort of how you pose the question and answer the question is also relevant for that. Now we've talked a lot about kind of inflation, Fed and so on. I mean, what else is on your mind? Is anything that worries you a lot that we haven't talked about, or your optimism that excites you that we haven't talked about. Well, had we had this conversation six or 12 months ago, I would have flagged and highlighted potential concerns about Fed independence. I think I'm much less worried about that now. I think Kevin Warsh is an excellent choice. In some ways, perhaps for the moment, perhaps the best choice the president could have made, I think He's been very clear in his first seven weeks that he realizes that the history books will judge him based upon whether or not he achieved and sustained price stability. And so whatever concern you might have had about the politicization of the Fed, I think is diminished. I guess I'll say something that's not particularly innovative or that you haven't heard, but I think it is relevant. And it's simply that.

All of us, you know, you and I have spent much of our professional career, you know, in a world that no longer exists. It was a world of hyper-globalization, of financial integration, of, you know, if you will, sort of a, we are the world approach to global macro and finance, and that world's gone on. Temco has highlighted the fact that we are now and will be for the foreseeable future in an era of fragmentation, fragmentation in global security alliances, trading arrangements, financial alliances. We're in a world of elevated ex-anti-geopolitical risk, and we've now seen an ex-post-kinetic geopolitical reality. It's a world that Mark Carney has referred to undergoing a rupture, not just a transition. And so there's a lot to worry about in that world. In particular, we can't really rely on the comfort of having seen in our professional careers prior cycles. However, I think we can get surprised along both dimensions. And I discussed this earlier in the interview. I mean, the conventional, almost unanimous view of elites and the financial press in April of 2025 was that this was perhaps not only something that would be disruptive, but in terms of ending the global system as we knew it, but it would also have very, very severe and adverse macro consequences.

And it did not. And so perhaps one thing we should also remember in an era of fragmentation and disruption is that capitalist economies are remarkably and perhaps surprisingly adaptable to new arrangements and conditions. And so although there's a lot to worry about in a world of fragmentation and rupture, we don't want to lose sight of the flexibility and adaptability of economies in that world. That's a fantastic point. I did want to round off the question I asked all of my guests, which is that we do have some young people listening to the podcast as well. Many of them are graduating soon. They're entering the job market. Many of them, when they speak to me at least, is they're quite gloomy about the job prospects. I mean, what advice do you give to youngsters? My advice would not to be gloomy if you look at your career over many decades. That's not saying that the a number of job interviews you have to go through to get your first job or where to get a job that you like there's going to we're going through a period now in particular in the US where vacancy rates are not elevated but I think for young people today there are very exciting opportunities in the sense that I think that this new technology for especially for workers who I think are educated coming out with good educations will complement their skills not necessarily

substitute for their their skills, and then for those entering the job market who are in a more vocational orientation, in the U.S. and in many economies, there is a structural shortage of skilled trade labor. In fact, just recently, META, the tech company has announced this program you may have heard of, but it's called something like the META Job Academy. And what they're doing is they're partnering with different groups to basically to start a trade apprenticeship program for electricians and plumbers and mechanics. They're covering all the tuition. They're providing a stipend. And there's a guaranteed job at the end of that. And so I think with a longer perspective, this is going to be an excited time full of opportunity, notwithstanding some of the concerns that you highlighted right now.

Just one more question. If people wanted to follow your views, is there a place that can go to see your legacy thinking? Yeah. So I'm not an early adopter, but I'm finally on LinkedIn and I post probably weekly on LinkedIn. When I started doing this at Pemko 20 years ago, I would do a longer form piece on a monthly basis. And now I post, for example, I posted when Kevin Warsh was confirmed and I'll post after Jackson Hole on PIMCO.com. That's sort of a longer piece. But now I'm posting basically weekly on LinkedIn, usually a chart with a couple of paragraphs or so of analysis. And so I think that's the best place to find me on LinkedIn weekly and then on PIMCO.com quarterly.

That's fantastic. I'll make sure to include links on the show notes there. So, Rich, it's fantastic to speak to you. I always enjoy your conversations with you and getting into your thinking. Thank you very much. Keep up the good work. Yeah, thank you. Thanks for listening to this episode. Please subscribe to the podcast show on Apple Spotify. You'll be able to listen to podcasts, leave a five-star rating, a nice comment, and let other people know about the show. We'll be very, very grateful. Sign up for our free newsletter at macrohype.com. MacroHive retains all ownership, title, rights and interests in this audio and video and all related content, including audio, transcripts, thumbnails and descriptions. You may share links to this publication and embed it using platform-native features. You must not, whether in whole or in part, download, copy, reproduce, redistribute, re-upload, edit, client or create derivative works from this content without the express written authority of MacroHive. This audio and video is made

available to you for general information purposes only and does not constitute the recommendation of any investment product to act or not to act in any way whatsoever and does not represent an offer, or the solicitation offered to buy or sell any securities, financial products or related financial services, or to adopt an investment strategy. To the fullest extent, permitted by applicable law, Makohive excludes all representations or warranties of any kind, including any implied, warranty of merchantability, satisfactory quality of fitness or particular purpose. To the maximum extent, permitted by applicable law, Makohive excludes all liability for any losses, whether direct or indirect that may be suffered as a result of reliance on this audio or video. Unless otherwise stated, the views, information or opinions expressed during this audio or video are solely those of the speaker and do not necessarily represent those of MacroHive or its staff. Copyright 2025, MacroHive Ltd. All rights reserved.